25900288 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSECHE202 - SAFE LABORATORY PRACTICES IN CHEMISTRY

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **5** questions

Each question carries **4** marks

1. What are the key components of an effective emergency response plan? [A] / [CO1]
2. Explain briefly the safe handling of toxic compounds. [U] / [CO2]
3. Why laboratory equipment safety is important? [U] / [CO3]
4. Explain the role of fire extinguishers. [K] / [CO4]
5. Explain why glassware is preferred in laboratory experiments compared to other materials. [A] / [CO1]
6. Explain the toxic effect of picric acid [U] / [CO2]
7. Explain the significance of fire safety symbols. [U] / [CO3]

[4x5 = 20]

Part B

Short Essay Type Questions

Answer any **5** questions

Each question carries **7** marks

8. Why must oxidizing and reducing agents be stored separately? Illustrate with an example [U] / [CO4]

9. Give a short essay on safe work procedures for handling hazardous chemicals. [U] / [CO3]

10. Discuss the general rules for disposal of chemical waste [K] / [CO2]

11. Compare the functions of respirators and masks in providing safety to laboratory workers. [U] / [CO1]

12. Explain the importance of using hazard pictograms in chemical labeling, with examples. [U] / [CO4]

13. How do we respond to chemical spill? [A] / [CO3]

14. Explain the significance of waste containers in maintaining safety during laboratory and industrial waste disposal. [U] / [CO2]

[7x5 = 35]

Part C

Essay Type Questions

Answer any **1** questions

Each question carries **15** marks

15. What systemic failures led to the Bhopal Gas Disaster, and how can the lessons learned be applied to enhance modern industrial safety and accountability? [An] / [CO3]

16. Explain the significance of Hazard (H) and Precautionary (P) statements in chemical safety [U] / [CO2]

[15x1 = 15]